

Data Collection and Preprocessing Phase

Team ID	MCA
Project Title	Strategic Employee Attrition Intelligence Dashboard
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Analyzes the employee attrition dataset to identify data quality issues, understand key employee and job-related factors, and prepare clean and reliable data for identifying the major drivers of employee turnover and developing the Strategic Employee Attrition Intelligence Dashboard.

Section	Description
Data Overview	The dataset contains 5,000 employee records and 29 attributes related to employee demographics, job details, compensation, workload, engagement, career growth, work arrangements, and attrition. The target variable is Attrition , which identifies whether an employee has left the organization.
Data Source	Employee Attrition dataset containing employee information such as age, gender, department, job role, monthly income, salary hike, years at company, overtime hours, manager support, engagement, work-life balance, performance, AI usage, perceived AI job risk, and attrition .
Data Quality Assessment	Check the dataset for missing values, duplicate employee records, invalid values, inconsistent categories, and incorrect data entries to ensure reliable analysis.
Data Cleaning	Handle missing values if identified, remove duplicate employee records, check invalid numerical values, and standardize categorical values such as department, job role, work mode, gender, and marital status .
Data Transformation	Use Power Query to filter and sort employee records, organize attrition categories, create useful calculated columns, and prepare the data for dashboard analysis.

Data Type Conversion	Verify and correct data types for columns such as employee ID, age, monthly income, salary hike percentage, years at company, overtime hours, training hours, engagement score, burnout score, and attrition.
Column Splitting and Merging	Split or merge columns where required to improve analysis. For example, employee/job-related information can be organized into suitable analytical categories for dashboard reporting.
Feature/Calculated Columns	Create analytical fields such as Attrition Flag, Experience Group, Income Group, Workload Category, Engagement Category, Burnout Category, and Career Growth Indicator to identify potential factors contributing to employee turnover.
Data Modeling	Establish relationships between relevant tables if additional tables are introduced. Create measures for Total Employees, Employees Who Left, Attrition Rate, Average Income, Average Engagement Score, Average Burnout Score, Average Overtime Hours, and Average Years Since Promotion.
Attrition Analysis Preparation	Prepare the dataset to analyze the relationship between employee attrition and factors such as compensation, workload, engagement, burnout, career growth, work-life balance, manager support, commute, work mode, and AI-related job risk.
Save Processed Data	Save the cleaned and transformed employee dataset as the final processed dataset for use in the Strategic Employee Attrition Intelligence Dashboard.